

STAT/MA 41600

In-Class Problem Set #2: August 28, 2026

1. Flip 100 (fair) coins in a row.
    - 1a. What is the probability that there are exactly 50 heads and 50 tails?
    - 1b. What is the probability that there are strictly more heads than tails?
  2. Roll a green 6-sided die and a red 6-sided die.
    - 2a. What is the probability that the sum is between 4 and 8 (inclusive)?
    - 2b. If  $X$  is the value on the green die and  $Y$  is the value on the red die, what is the probability that  $|X - Y| \leq 1$ ?
  3. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space contains all ordered 3-tuples whose elements are  $r, o, y, g, b, p$ . Assume that all such outcomes are equally likely.
    - 3a. What is the probability that a randomly selected bear family does not contain any red bears?
    - 3b. We say that a bear family is “primary” if it has a red, a blue, and a yellow bear (in any order). What is the probability that a randomly selected bear family is *not* primary?
    - 3c. What is the probability that, in a randomly selected bear family, all three bears have the same color?
    - 3d. What is the probability that, in a randomly selected bear family, all three bears have different colors?
  4. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy. If we pick a week randomly, the sample space contains all ordered 5-tuples whose elements are  $s, c$ , or  $r$ . One outcome in the sample space is  $(s, s, c, r, s)$ . Another outcome is  $(c, r, c, s, s)$ . Assume that all of the weekly outcomes are equally likely.

For each value of  $k$  from 0 to 5 (inclusive), find the probability that there are exactly  $k$  sunny days. Make sure that your six values sum to 1.
  5. An experiment measures how many raindrops fall near a detection device during a fixed interval of time. Suppose that, for each nonnegative integer  $x$ , the probability of exactly  $x$  raindrops being detected is  $\frac{3^x e^{-3}}{x!}$ . Find the probability that 2 or more raindrops are detected.
- Mind bender (not required).** Returning to the situation in Question 1, what is the probability that there are an even number of heads?